LECTURE SENSE V2

Local-First Course AI Assistant

Senior Project 2

TEAM KGB

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What is SP2?

- One reusable app, many downloadable course packs.
- Teacher prepares a pack from course PDFs, slides, textbooks, and later audio/video.
- Student imports the pack and chats with a local AI assistant.
- The assistant answers from the installed course pack, not from the open internet or generic model memory.
- Core priority: reliable course-grounded help, privacy, offline/local use, and simple teacher-student distribution.

Why do we Need it?

Student pain points:

- Course knowledge is scattered across PDFs, slides, Teams/LMS folders, recordings, notes, and announcements.
- Students often do not know where in the course material an answer comes from.
- Generic ChatGPT can be helpful but may answer from general knowledge instead of the actual course.
- Students need explanations, but also need page/source references they can verify.
- Students may worry about uploading class materials, personal notes, or assignments to cloud AI tools.
- Some students have inconsistent access to paid tools, internet, or high usage limits.
- When AI is wrong, it often sounds confident, which makes studying riskier.

Why do we Need it?

Teacher pain points:

- Teachers repeatedly answer the same conceptual or logistical questions.
- They want students to get help, but not random answers that conflict with the course.
- They need control over what material the AI can use.
- They do not want to rebuild the course platform or replace Teams/LMS.
- They worry about privacy, copyright, academic integrity, and student overdependence.
- Existing AI tools often put the setup burden on each student instead of letting the teacher prepare one shared course knowledge pack.
- Teachers need a practical workflow: prepare once, share through existing channe

How does it solve the problem?

- Teacher uploads course material into the builder.
- SP2 extracts text, chunks it, adds metadata, creates embeddings, and exports a .zip pack.
- Teacher shares the pack through Teams, Drive, LMS, or another normal channel.
- Student imports the pack into the SP2 app.
- The app creates local SQLite metadata and a local LanceDB search index.
- Student asks questions.
- SP2 embeds the question, retrieves relevant chunks from the selected course pack, and sends only that context to the local model.
- Answer includes retrieved references where possible.
- If retrieval is weak, SP2 returns: “This is not in the database.”

Differences from SP1

FEATURE	SP1	SP2
AI dependency	External API required	Local model server
Embeddings	Cloud embedding API	Local sentence-transformers
Setup effort	Manual DB + API key setup	docker compose up --build
Internet reliance	Core AI depends on cloud	Core AI can run locally
Hosting type	Cloud-assisted web app	Local use
Cost after setup	Per-query cloud cost	None
Product identity	Lecture workspace	Private AI TA

Existing solutions

FEATURE	LectureSenseV2	ChatGPT	Google NotebookLM	Open NotebookLM	MS Copilot	Canvas AI
Runs locally (no internet)	✓	✗	✗	⚠ Partial	✗	✗
Course-specific knowledge	✓	✗	✓	✓	✗	⚠ Partial
Professor controls content	✓	✗	✗	⚠ Partial	✗	⚠ Partial
Data stays on device	✓	✗	✗	⚠ Depends	✗	✗
Free after setup	✓	✗ Subscription	⚠ Free tier	✓	✗ Subscription	✗ Institutional
Apple Silicon optimized	✓	✗	✗	✗	✗	✗
PDF + audio ingestion	✓	⚠ Limited	✓	✓	⚠ Limited	✗
Page-level citations	✓	✗	✓	✓	✗	✗
No account required	✓	✗	✗	✓	✗	✗
Open source	✓	✗	✗	✓	✗	✗
Portable course pack format	✓	✗	✗	✗	✗	✗

Apple Memory vs Qwen Local Models

A practical placement chart for SP2-style local RAG. X-axis is recommended unified memory. Y-axis is the Apple Silicon tier that should feel reasonable rather than merely possible.

Legend

- Qwen 2.5 Instruct
- Qwen 3 comparable size

How To Read It

Points farther right need more unified memory. Points higher on the chart benefit from faster chip tiers and memory bandwidth.

A model can often run below its plotted tier, but the plotted point is the smoother target for a student-facing local assistant.

Plotted Models

PAIR	QWEN 2.5	QWEN 3
tiny	`1.5B Q4`	`1.7B Q4`
small	`3B Q4`	`4B Q4`
daily	`7B Q5`	`8B Q5`
strong	`14B Q4`	`14B Q4`

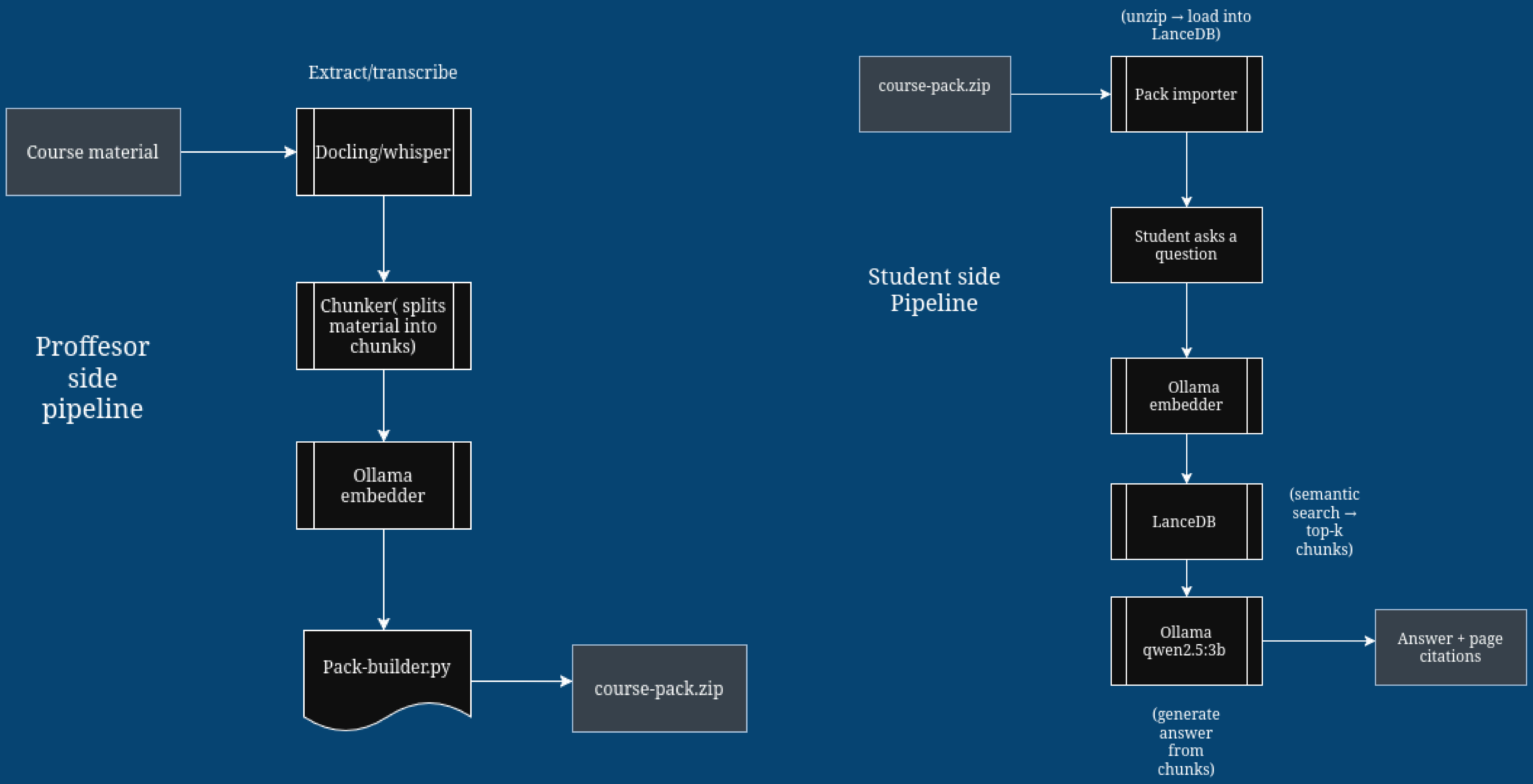
8GB
Use 1.5B-4B Q4 class models. Keep context short.

16GB
Best default zone for 7B/8B Q4-Q5.

24GB
Good target for 12B/14B Q4 models.

32GB
Headroom for 14B Q5 or 32B Q4 experiments.

Pipelines


Extract/transcribe
Course material
Docling/whisper
Chunker(splits material into chunks)
Proffesor side pipeline
Ollama embedder
Pack-builder.py
course-pack.zip
(unzip → load into LanceDB)
course-pack.zip
Pack importer
Student side Pipeline
Student asks a question
Ollama embedder
LanceDB
(semantic search → top-k chunks)
Ollama qwen2.5:3b
Answer + page citations
(generate answer from chunks)

Technical stack

Backend:

FastAPI

Fast API

Python

LanceDB

LanceDB

SQLite

SQLITE

docker

Frontend:

TS TypeScript

Local LLM:

OLLAMA

Qwen3

LectureSenseV2

COURSE PACKS

- Operating Systems
  CS301 · 3 lectures
- Data Structures
  CS201 · 5 lectures
- Computer Networks
  CS401 · 2 lectures

+ Import Course Pack

Operating Systems
CS301 · 3 lectures · all-minilm embeddings

Local · qwen2.5:3b

What are the four deadlock conditions?

You

SP

Based on the course materials, deadlock occurs when a set of threads are each waiting for a resource held by another thread in the set, creating a circular dependency that none can escape.

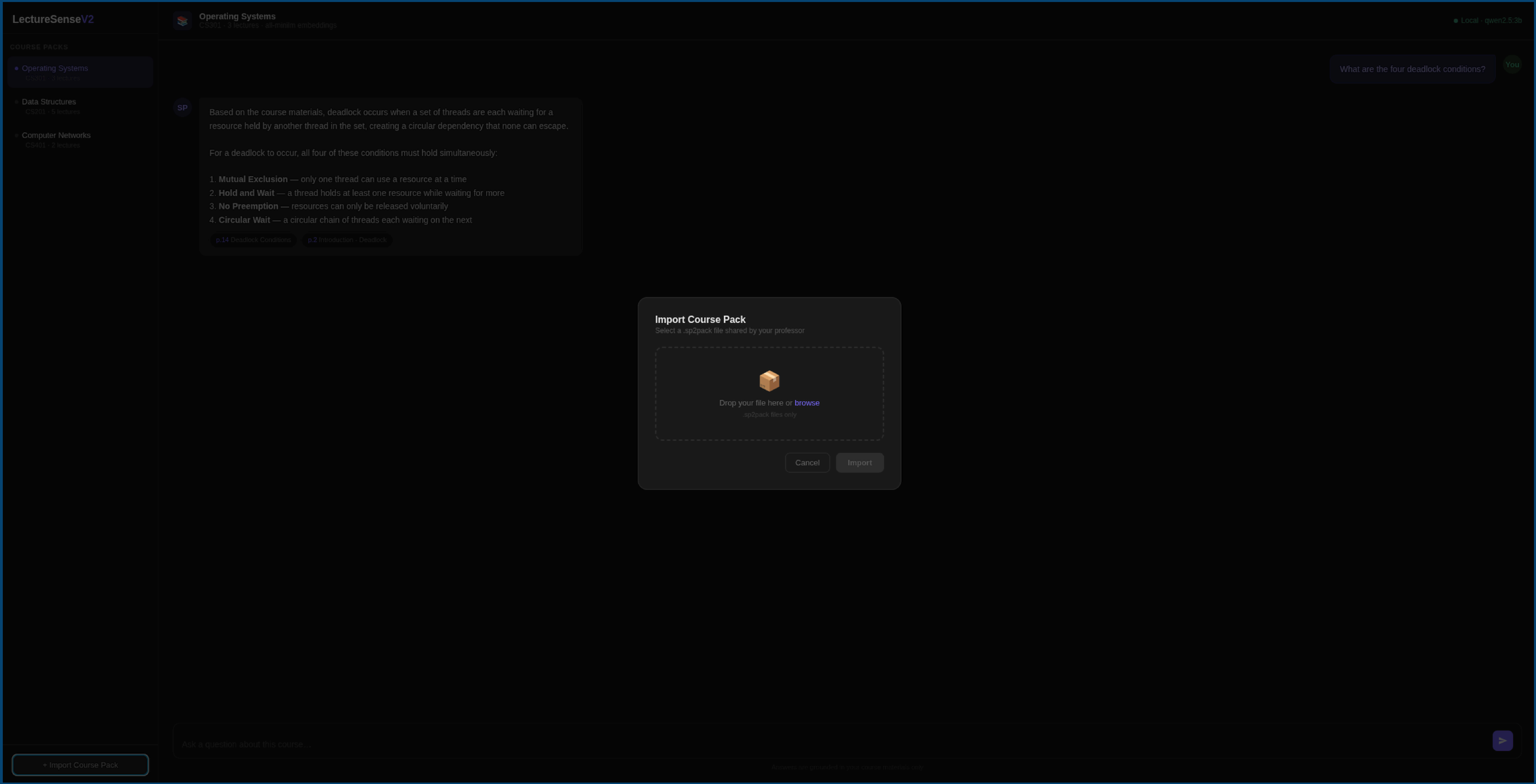
For a deadlock to occur, all four of these conditions must hold simultaneously:

1. **Mutual Exclusion** — only one thread can use a resource at a time
2. **Hold and Wait** — a thread holds at least one resource while waiting for more
3. **No Preemption** — resources can only be released voluntarily
4. **Circular Wait** — a circular chain of threads each waiting on the next

p.14 Deadlock Conditions   p.2 Introduction - Deadlock

Ask a question about this course...

Answers are grounded in your course materials only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CS301 · 3 lectures
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Import Course Pack
Select a .sp2pack file shared by your professor
Drop your file here or browse
.sp2pack files only
Cancel
Import
Ask a question about this course...
Answers are grounded in your course materials only